
Notes for ‘Maxima and minima’

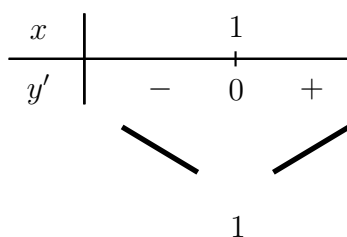
Important Ideas and Useful Facts:

- (i) **Extreme values:** A function f has a *global maximum* at c if $f(c) \geq f(x)$ for all x for which f is sensibly defined, in which case $f(c)$ is called the *maximum value* of f . Similar considerations apply to minimum values, by replacing $\geq$ by $\leq$. The maximum and minimum values are called *extreme values* of f .
- (ii) **Local Maxima and Minima:** A function f has a *local maximum* at c if $f(c) \geq f(x)$ for all x near c . Similar considerations apply to minima, by replacing $\geq$ by $\leq$. If f has a local or global maximum or minimum at c and f is differentiable at c then $f'(c) = 0$ (because the tangent line at $x = c$ has to be horizontal). Local maxima and minima are also called *local extrema*.
- (iii) **Turning points:** A *turning point* of a continuous curve is where the curve changes from being increasing to being decreasing, or from being decreasing to increasing. In the first case, the function has a local (and possibly global) maximum at the turning point. In the second case, the function has a local (and possibly global) minimum at the turning point.

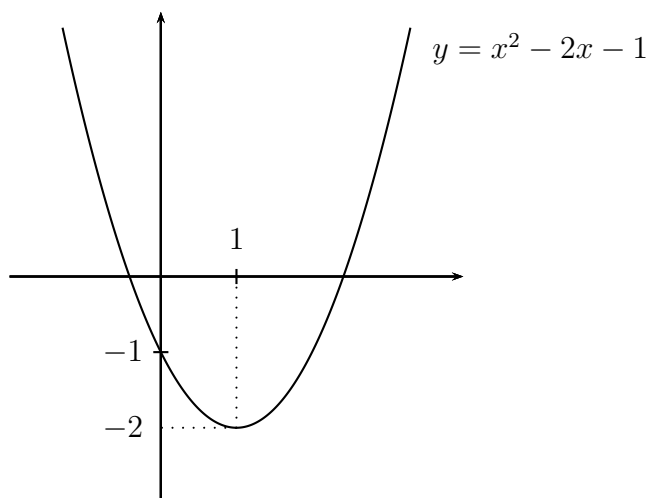
If the function is differentiable at the turning point then the derivative is zero at the turning point. Thus, turning points typically can be gleaned from the sign diagram for the derivative.
- (iv) **Critical Points:** We call the ordered pair $(c, f(c))$ a *critical point* for (the graph of) the function f if either $f'(c) = 0$ or $f'(c)$ does not exist, in which case we call $f(c)$ a *critical value*. Thus maxima and minima (local or global) are critical values (but not necessarily conversely). It is common language also to say that c is a *critical point* for f if $(c, f(c))$ is a critical point for the graph of f . Hence the problem of finding local maxima and minima may be reduced to finding critical points and inspecting critical values.
- (v) **The Closed Interval Method:** To find the absolute maximum and minimum values of a continuous function f defined for x satisfying $a \leq x \leq b$ (where a and b are fixed), it suffices to consider the largest and smallest values of $f(c)$ for $c = a$, $c = b$ and c such that $a < c < b$ and $f'(c) = 0$ or undefined.

Examples:

1. Consider the function $y = f(x) = x^2 - 2x - 1$. Then $y' = 2x - 2 = 2(x - 1)$, so that $y' = 0$ when $x = 1$, and we get the following sign diagram:



The graph is a parabola and the turning point is $(1, -2)$. The value $y = -2$ is the global minimum.



2. Find all local and global extrema for the cubic function

$$y = f(x) = x^3 - 6x^2 + 9x + 1.$$

over (a) the entire real line $\mathbb{R}$, and (b) over the interval $[0, 5]$.

Solution: For part (a), observe that

$$y' = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3),$$

so that $y' = 0$ when $x = 1, 3$ and we get the following sign diagram:

x		1		3		
y'		+	0	-	0	+
						

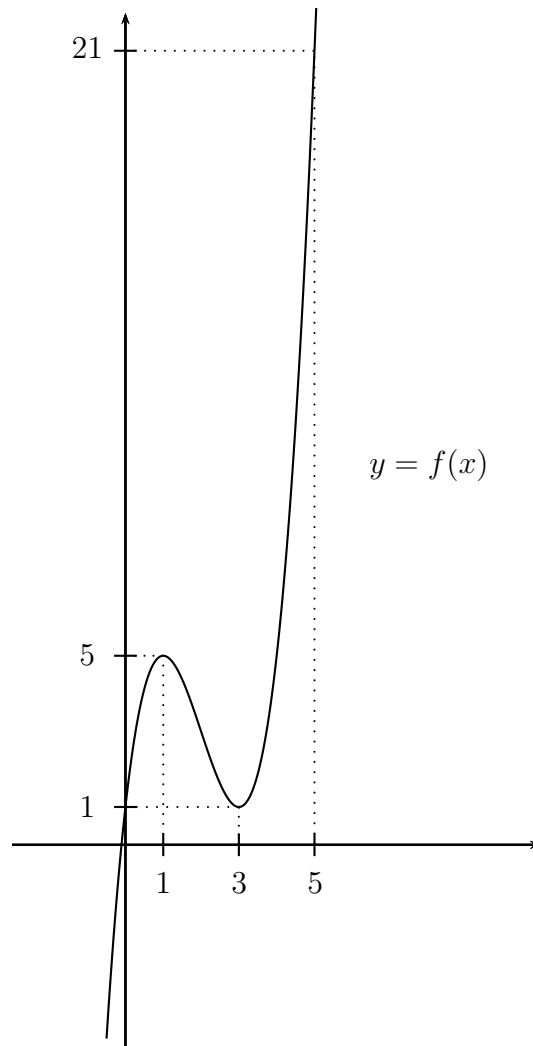
The turning points occur at $x = 1$ and $x = 3$ producing a local maximum $f(1) = 5$ and a local minimum $f(3) = 1$. Neither of these local extrema can be global, because the cubic takes arbitrarily large positive and negative values over the entire real line (as the rule for the function is dominated by x^3 for large inputs x).

For part (b), we apply the closed interval method and only need to check values of the function at the endpoints of the interval $[0, 5]$ and where the derivative is zero. Thus, utilising values obtained above for $x = 1$ and $x = 3$, we only need to compare the following:

$$f(0) = 1, \quad f(1) = 5, \quad f(3) = 1, \quad f(5) = 21.$$

The smallest value is 1, which is therefore the global minimum. The largest value is 21, which is therefore the global maximum.

Note that we can draw these conclusions without knowing what the graph looks like. In fact, the curve has the following appearance, consistent with the information in the sign diagram for the derivative:



3. Find the global maximum and minimum of the function f where

$$f(x) = |x - 1| + 3 ,$$

for $-2 \leq x \leq 5$.

Solution: Observe first that, if $x \geq 1$, then

$$f(x) = |x - 1| + 3 = x - 1 + 3 = x + 2 ,$$

and if $x < 1$ then

$$f(x) = |x - 1| + 3 = 1 - x + 3 = 4 - x .$$

Hence the rule for f can be expressed as follows:

$$f(x) = \begin{cases} x + 2 & \text{if } x \geq 1 \\ 4 - x & \text{if } x < 1 . \end{cases}$$

The graph is the result of joining two line segments of slopes 1 and -1 together at the point $(1, 3)$. No unique tangent line to the curve exists at this point, and the derivative

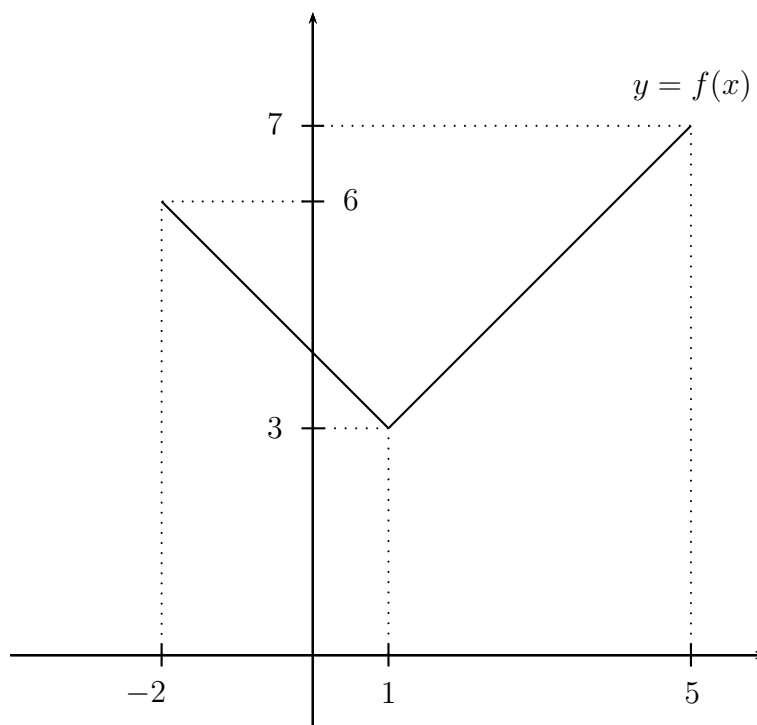
does not exist for $x = 1$. Nevertheless, the derivative exists for $x \neq 1$ and can be expressed as follows:

$$f'(x) = \begin{cases} 1 & \text{if } x > 1 \\ -1 & \text{if } x < 1 \end{cases}.$$

Note that the derivative is nonzero for all $x \neq 1$. Hence, applying the closed interval method, it suffices to compare the following values:

$$f(-2) = |-2 - 1| + 3 = 6, \quad f(1) = |1 - 1| + 3 = 3, \quad f(5) = |5 - 1| + 3 = 7.$$

The smallest value is 3, which is therefore the global minimum. The largest value is 7, which is therefore the global maximum. Note that these values could be gleaned directly from the graph, which is shown below.



4. Find the global maximum and minimum of the function f where

$$f(x) = |x - 2| + |x + 1|,$$

for $-3 \leq x \leq 3$.

Solution: Observe first that, if $x \geq 2$, then

$$f(x) = |x - 2| + |x + 1| = x - 2 + x + 1 = 2x - 1.$$

If $-1 \leq x < 2$ then

$$f(x) = |x - 2| + |x + 1| = 2 - x + x + 1 = 3.$$

If $x < -1$ then

$$f(x) = |x - 2| + |x + 1| = 2 - x - x - 1 = 1 - 2x.$$

Hence the rule for f can be expressed as follows:

$$f(x) = \begin{cases} 2x - 1 & \text{if } x \geq 2 \\ 3 & \text{if } -1 \leq x < 2 \\ 1 - 2x & \text{if } x < -1. \end{cases}$$

The graph is the result of joining together three line segments of slopes 2, 0 and -2 at the points $(2, 3)$ and $(-1, 3)$, No unique tangent line to the curve exists at these points, and the derivative does not exist for $x = -1$ or for $x = 2$. Nevertheless, the derivative exists for $x \neq -1, 2$ and can be expressed as follows:

$$f'(x) = \begin{cases} 2 & \text{if } x > 2 \\ 0 & \text{if } -1 < x < 2 \\ -2 & \text{if } x < -1. \end{cases}$$

The derivative is zero or undefined therefore for all x in the interval $[-1, 2]$. Applying the closed interval method, it suffices to look at the value $f(x)$ for such x and also for the endpoints, when $x = -3$ and $x = 3$. For all $x \in [-1, 2]$ we have $f(x) = 3$. At the endpoints, we have

$$f(-3) = |-3 - 2| + |-3 + 1| = 5 + 2 = 7$$

and

$$f(3) = |3 - 2| + |3 + 1| = 1 + 4 = 5.$$

The smallest value is 3, which is therefore the global minimum. The largest value is 7, which is therefore the global maximum. Note, again, that these values could be gleaned directly from the graph, which is shown below.

